Artificial Neural Networks
Assignment 3

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Task 0: Understanding the Working Example

1. Array Sizes

Feature map: (3, 3)
Activated: (3, 3)
Pooled: (2, 2)
Flattened: (4,)
Dense output:(10,)
Output layer: (10,)

2. Expected Values and Output Layer in 1st Epoch

Feature Map (Epoch 1):

```
First Epoch Feature Map:
 [[20.72376544 26.95352407 18.15537627]
 [ 8.80897988 17.52802535 13.67677825]
 [ 8.19087764 17.92398833 15.82210265]]
First Epoch Output Layer:
 [5.93947968e-56 1.76458568e-57 1.11686461e-63 3.08510861e-66
 2.42206080e-83 7.06406960e-83 3.46802839e-72 1.60312579e-90
 1.00000000e+00 8.01713646e-64]
```

3. Expected Values and Output Layer in Last Epoch

Feature Map (Epoch 200):

```
Last Epoch Feature Map:
 [[4481.52376544 5987.15352407 4367.75537627]
 [2571.30897988 4115.82802535 3538.87677825]
 [1811.39087764 3831.72398833 3756.22210265]]
Last Epoch Output Layer:
 [0. 0. 0. 0. 0. 0. 0. 1. 0. 0.]
```

4. Why Add 1e-9 to Loss Function

1. It prevents taking the logarithm of zero, which is undefined, avoiding errors.
2. Softmax can produce outputs very close to zero, and the logarithm of such tiny values

can cause computational instability.
3. Adding 1e-9 ensures the logarithm's input is always slightly positive, maintaining numerical stability during training.

5. Average Loss Plot for Different Learning Rates

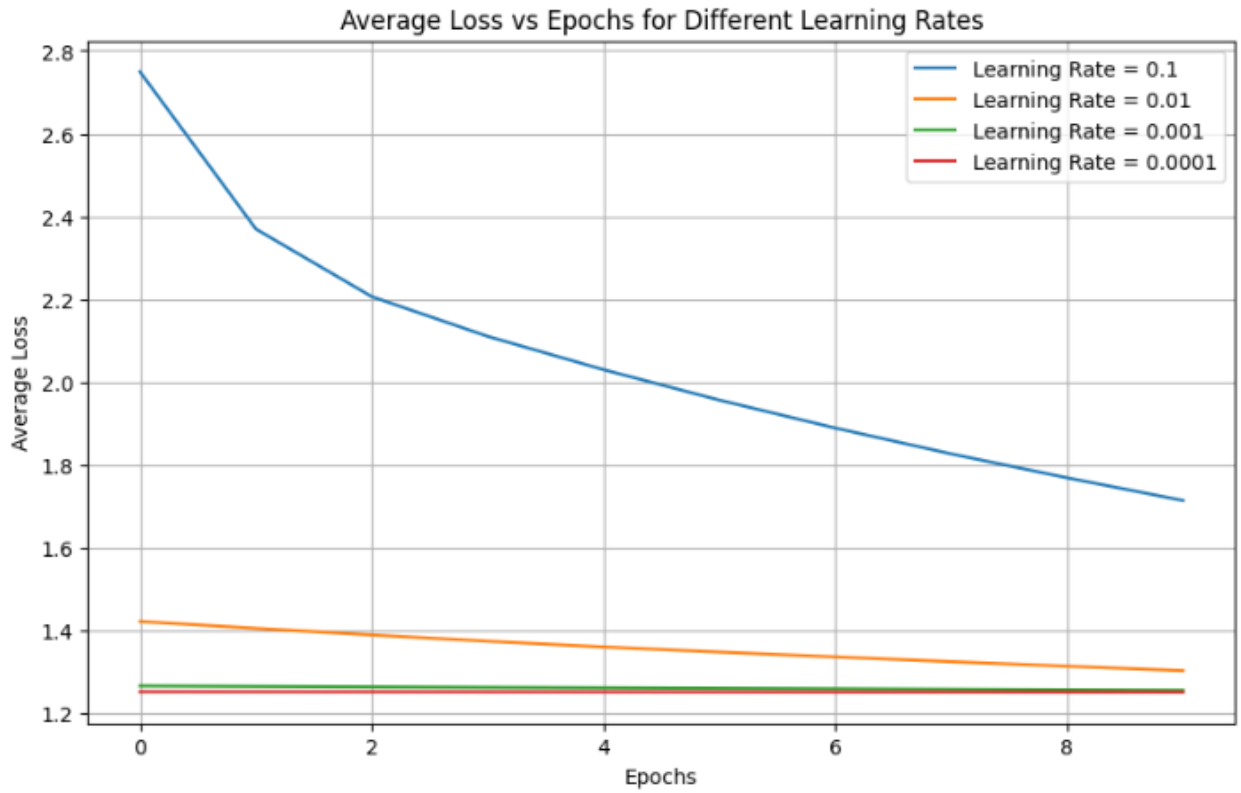

Figure 1: Average Loss vs Epochs for various learning rates

6. Loss at 200th Epoch for Different Kernel Sizes

Kernel Size	Loss at 200th Epoch
2x2	2.3476
3x3	0.0453
5x5	0.0033

7. Feature Map Images at Various Epochs

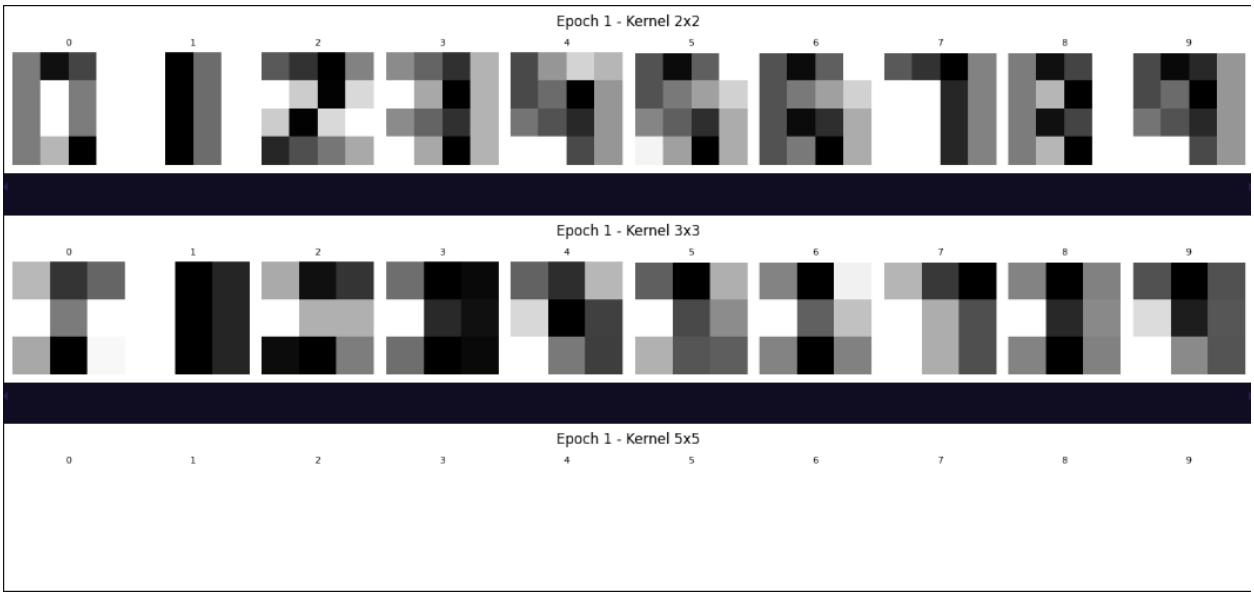

Figure 2: Feature Map at Epoch 1

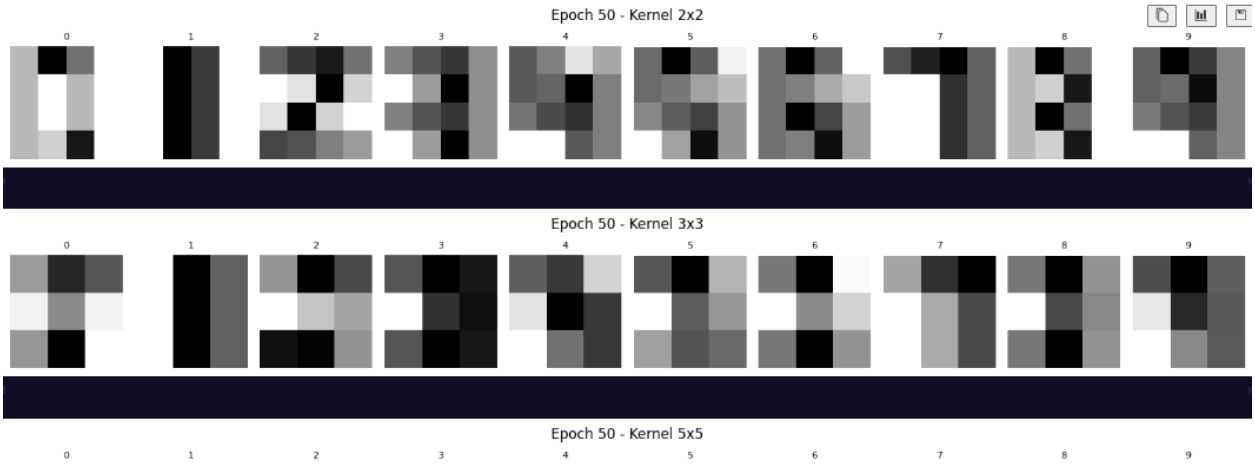

Figure 3: Feature Map at Epoch 50

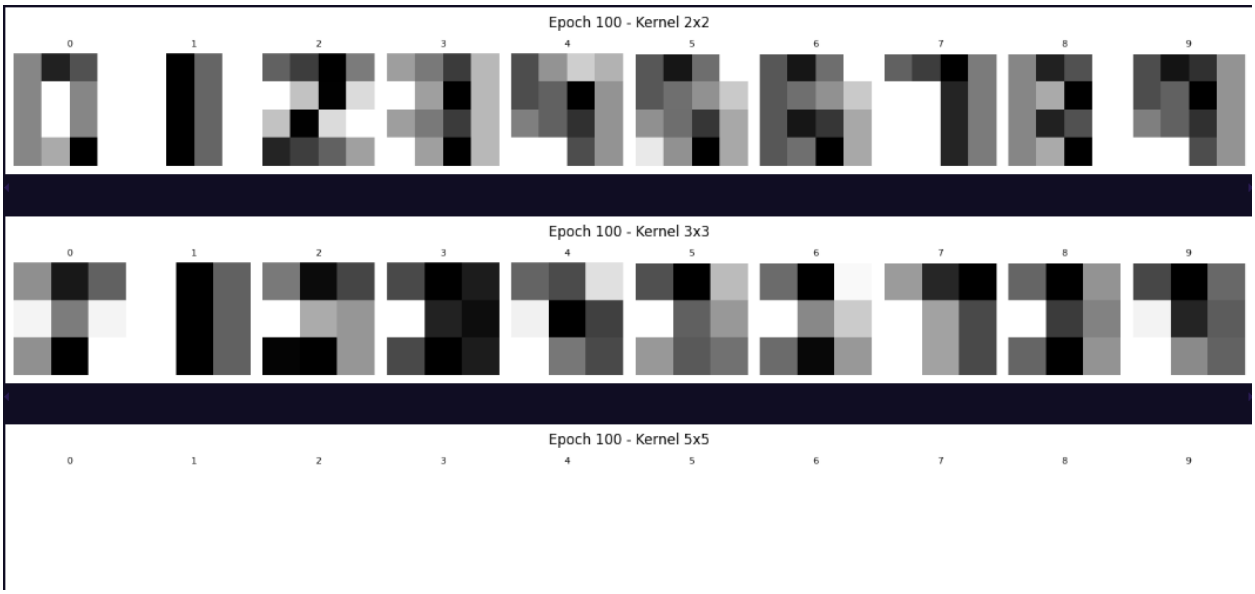

Figure 4: Feature Map at Epoch 100

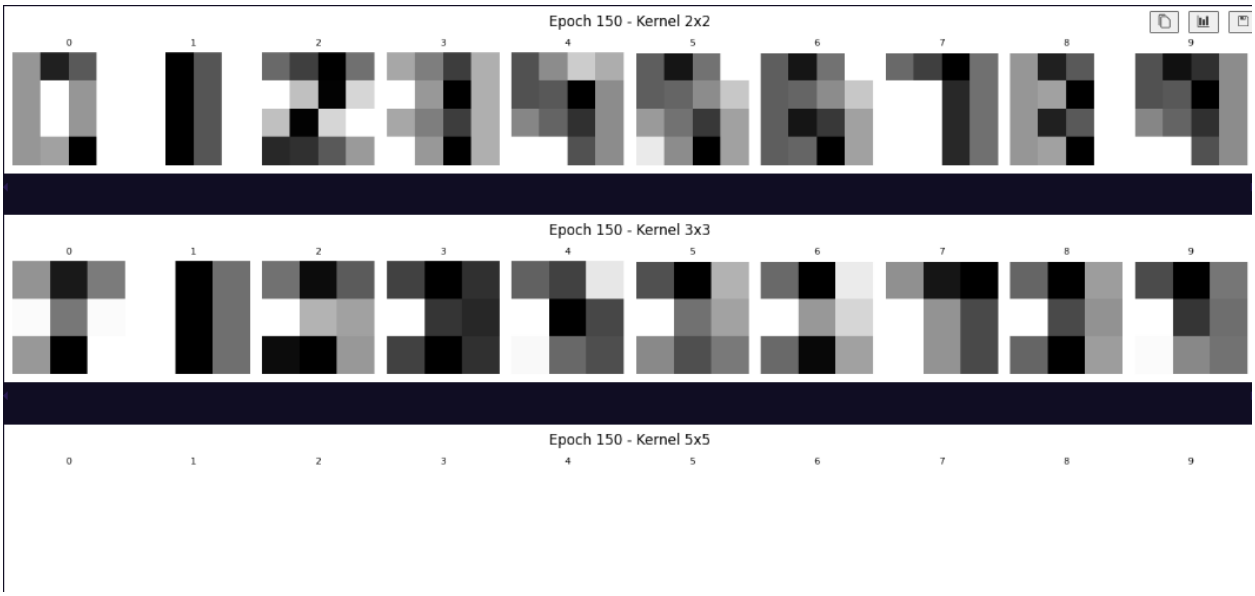

Figure 5: Feature Map at Epoch 150

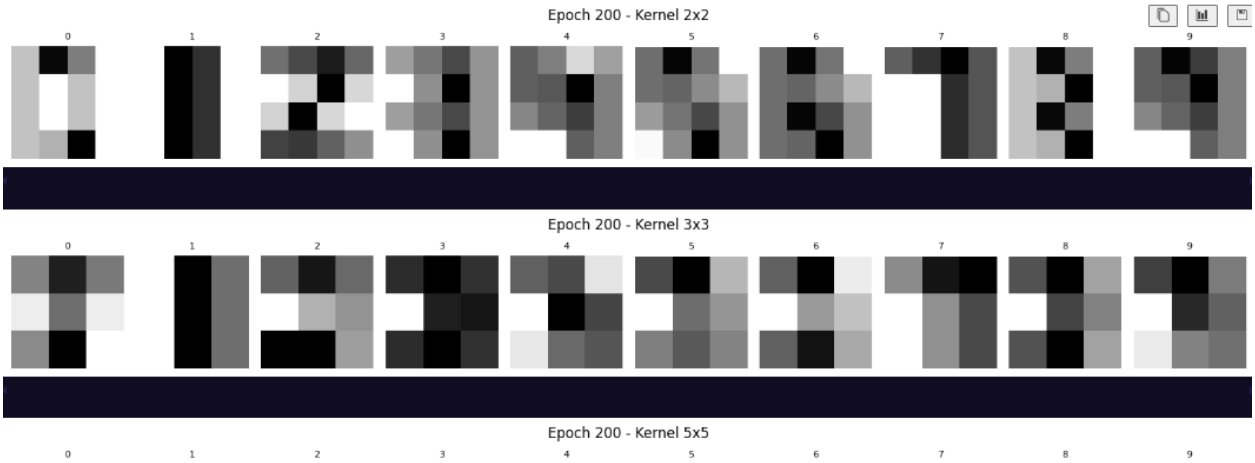

Figure 6: Feature Map at Epoch 200

Explanation:

The reason the 5x5 kernels produce no visible output and the model's accuracy worsens over time is primarily due to the way the kernels are being updated. Instead of learning through a loss-driven optimization process like backpropagation, the kernels are randomly perturbed with Gaussian noise at each epoch. This causes the filters—especially larger ones like 5x5, which have more parameters—to drift away from meaningful patterns and potentially collapse into near-zero or uniform values, resulting in blank or uninformative feature maps. Additionally, applying large kernels to small input images (like 4x4 or 5x5 digit samples) reduces the effective output size, sometimes to a single pixel or nothing usable. Over time, this uncontrolled and noise-based update mechanism destroys any initially useful filters, leading to significant accuracy degradation and the disappearance of distinct features in visualizations.

Task 1: Creating and Evaluating Test Cases

1. Predicted vs Actual with Noise

It holds when there is no noise added, but performs terribly when noise is added, it locks in on one single number (9) for each prediction

2. Always Correct Predictions

The only number it correctly identified with noise was 9

3. Always Incorrect Predictions

From 1 to 8

4. Confusion Matrix

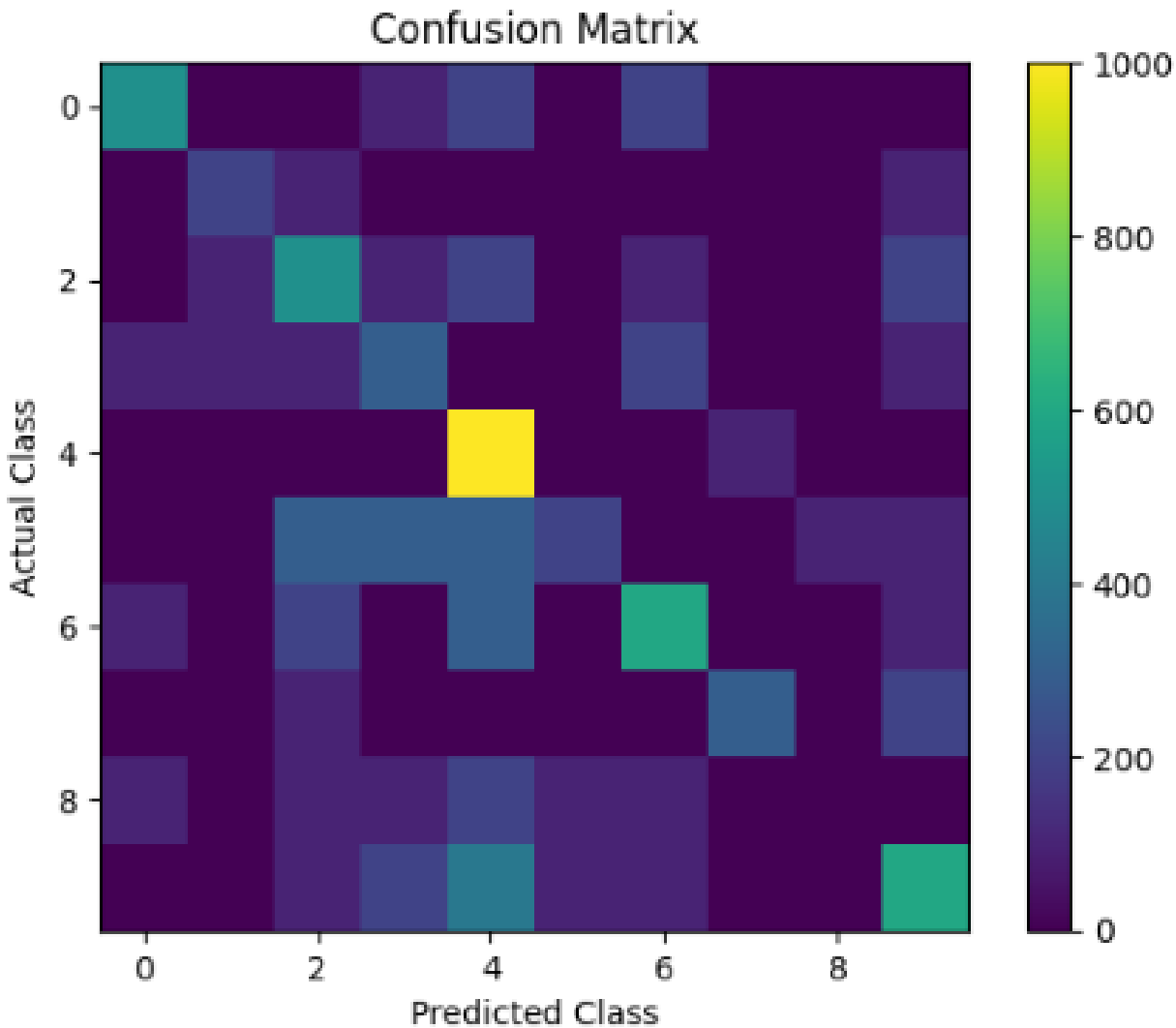

Figure 7: Confusion Matrix Visualization

5. TP, FP, FN, TN Table

Metric	Value
True Positive (TP)	500
False Negative (FN)	500
False Positive (FP)	300
True Negative (TN)	8700

6. Evaluation Metrics

Metric	Value
Precision	0.625
Recall	0.5
Specificity	0.97
Negative Prediction	0.95
Accuracy	0.92
F1 Score	0.56

7. Noise Impact on Metrics

Noise Level	Accuracy	Precision	Recall
0	0.8230	0.1150	0.1150
1	0.8208	0.1040	0.1040
2	0.8192	0.0960	0.0960
3	0.8188	0.0940	0.0940
4	0.8180	0.0900	0.0900

Figure 8: Noise Impact on Metrics

Task 2: Adding an Additional Kernel

Evaluation with Additional Kernel

```
Noise Level 0: Accuracy=0.8192, Precision=0.0960, Recall=0.0960
Noise Level 1: Accuracy=0.8230, Precision=0.1150, Recall=0.1150
Noise Level 2: Accuracy=0.8178, Precision=0.0890, Recall=0.0890
Noise Level 3: Accuracy=0.8196, Precision=0.0980, Recall=0.0980
Noise Level 4: Accuracy=0.8202, Precision=0.1010, Recall=0.1010
```

Figure 9: By adding an additional kernel